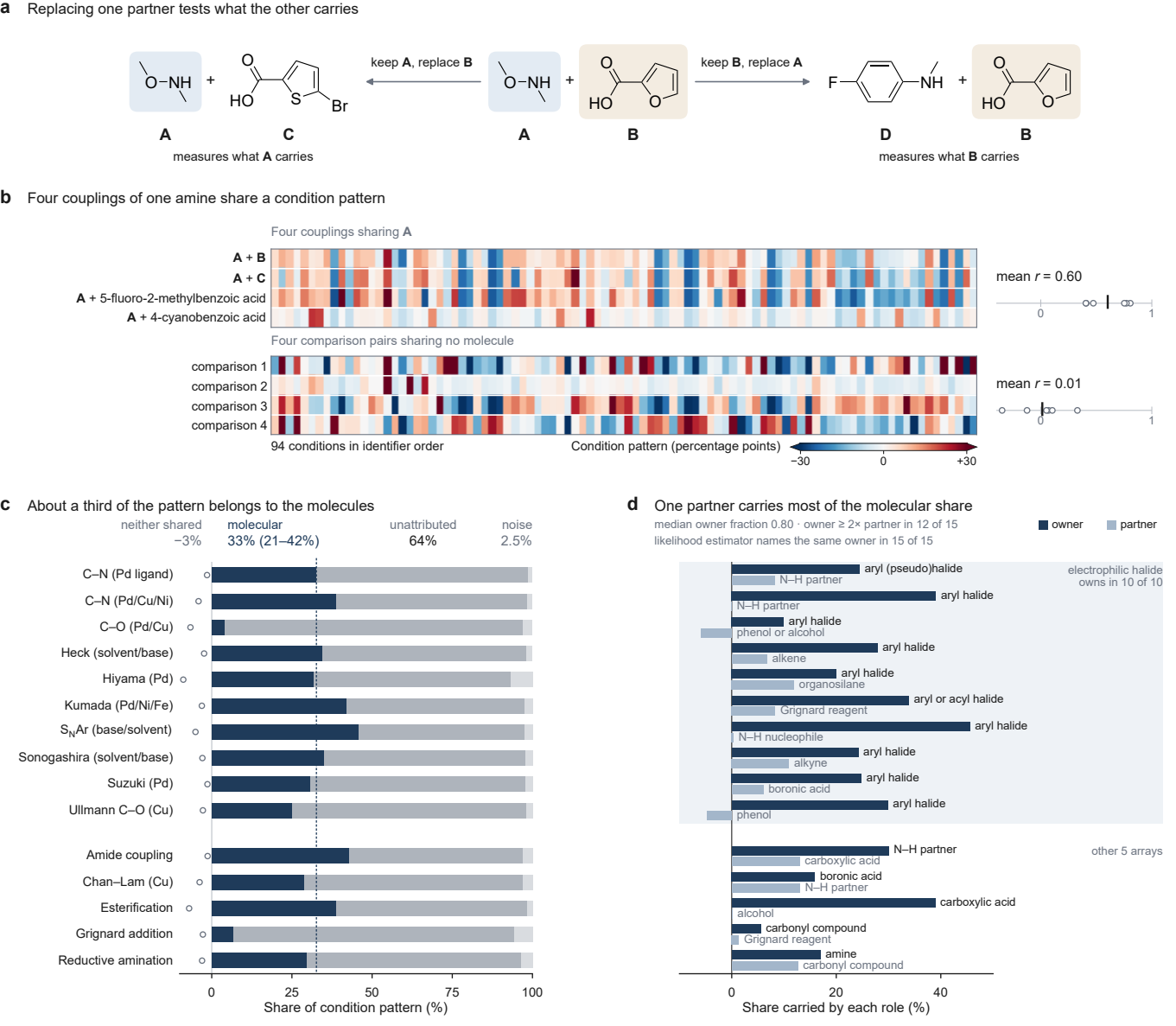

a
Replacing one partner tests what the other carries
keep A, replace B
keep B, replace A
A
C
A
B
D
B
measures what A carries
measures what B carries
b
Four couplings of one amine share a condition pattern
Four couplings sharing A
A + B
A + C
A + 5-fluoro-2-methylbenzoic acid
A + 4-cyanobenzoic acid
mean r = 0.60
Four comparison pairs sharing no molecule
comparison 1
comparison 2
comparison 3
comparison 4
mean r = 0.01
94 conditions in identifier order
Condition pattern (percentage points)
−30
0
+30
c
About a third of the pattern belongs to the molecules
neither shared
−3%
molecular
33% (21–42%)
unattributed
64%
noise
2.5%
C–N (Pd ligand)
C–N (Pd/Cu/Ni)
C–O (Pd/Cu)
Heck (solvent/base)
Hiyama (Pd)
Kumada (Pd/Ni/Fe)
SNAr (base/solvent)
Sonogashira (solvent/base)
Suzuki (Pd)
Ullmann C–O (Cu)
Amide coupling
Chan–Lam (Cu)
Esterification
Grignard addition
Reductive amination
0
25
50
75
100
Share of condition pattern (%)
d
One partner carries most of the molecular share
median owner fraction 0.80 · owner ≥ 2× partner in 12 of 15
likelihood estimator names the same owner in 15 of 15
owner
partner
electrophilic halide
owns in 10 of 10
aryl (pseudo)halide
N–H partner
aryl halide
N–H partner
aryl halide
phenol or alcohol
aryl halide
alkene
aryl halide
organosilane
aryl or acyl halide
Grignard reagent
aryl halide
N–H nucleophile
aryl halide
alkyne
aryl halide
boronic acid
aryl halide
phenol
other 5 arrays
N–H partner
carboxylic acid
boronic acid
N–H partner
carboxylic acid
alcohol
carbonyl compound
Grignard reagent
amine
carbonyl compound
0
20
40
Share carried by each role (%)